LAB RECORD

C Programming Lab Record

A complete collection of 28 C programming experiments — handwritten in a lab record book, carefully digitized into clean, copy-ready code.

28 Experiments | Language **C** | Compiled & verified with **GCC**

akhil-tg/01012008 · Full printable edition

Index

01 C program to print the size of different data types

02 Perimeter and area of a triangle

03 Area and circumference of a circle

04 Convert Fahrenheit to Celsius

05 Simple interest

06 Swap two numbers without using a temporary variable

07 Swap two numbers using a temporary variable

08 Check a > b using the conditional operator

09 Arithmetic and relational operations using short-name method

10 Evaluate the expression (a+b)*c

11 Equivalence of two numbers

12 Largest of two numbers

13 Maximum of three numbers

14 Smallest of two numbers

15 Energy bill using else-if ladder

16 Leap year or not

17 Odd or even

18 Gross salary of an employee

19 Average of six subjects with division classification

20 Day of a week using switch case

21 Menu driven arithmetic calculations using switch case

22 Display star pattern

23 Display number triangle pattern

24 Print Floyd triangle

25 Find the sum of array elements

26 Find the largest element in an array

27 Display a matrix

28 Addition of two matrices

01 C program to print the size of different data types

AIM To write a C program to print the size of different data types.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    clrscr();
    printf(" SIZE OF DATA TYPES : \n\n");
    printf(" Size of int = %d bytes\n", sizeof(int));
    printf(" Size of char = %d bytes\n", sizeof(char));
    printf(" Size of float = %d bytes\n", sizeof(float));
    printf(" Size of double = %d bytes\n", sizeof(double));
    getch();
}
```

Sample Run

Input (none)

Output Size of int = 4 bytes | Size of char = 1 byte | Size of float = 4 bytes | Size of double = 8 bytes

02 Perimeter and area of a triangle

AIM To find the perimeter and area of a triangle.

```c
#include<stdio.h>
#include<conio.h>
#include<math.h>
void main()
{
    float a, b, c, s, area, perimeter;
    clrscr();

    printf(" Enter three sides of triangle : ");
    scanf("%f %f %f", &a, &b, &c);

    perimeter = a + b + c;
    s = perimeter / 2;
    area = sqrt(s * (s - a) * (s - b) * (s - c));

    printf(" Perimeter = %f", perimeter);
    printf("\n Area = %f", area);

    getch();
}
```

Sample Run

Input 3 4 5

Output Perimeter = 12.000000 | Area = 6.000000

03 Area and circumference of a circle

AIM To find the area and circumference of a circle.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    float r, area, circumference;
    clrscr();

    printf(" Enter radius : ");
    scanf("%f", &r);

    area = 3.14 * r * r;
    circumference = 2 * 3.14 * r;

    printf(" Area = %f", area);
    printf("\n Circumference = %f", circumference);

    getch();
}
```

Sample Run

Input 7

Output Area = 153.860001 | Circumference = 43.959999

04 Convert Fahrenheit to Celsius

AIM To convert temperature in Fahrenheit to Celsius.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    float f, c;
    clrscr();

    printf(" Enter temperature in Fahrenheit : ");
    scanf("%f", &f);

    c = (f - 32) * 5 / 9;

    printf(" Temperature in Celsius = %f", c);

    getch();
}
```

Sample Run

Input 98.6

Output Temperature in Celsius = 37.000000

05 Simple interest

AIM To find the Simple Interest.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    float p, r, t, si;
    clrscr();

    printf(" Enter principal, rate and time : ");
    scanf("%f %f %f", &p, &r, &t);

    si = (p * r * t) / 100;

    printf(" Simple Interest = %f", si);

    getch();
}
```

Sample Run

Input 1000 5 2

Output Simple Interest = 100.000000

06 Swap two numbers without using a temporary variable

AIM To swap two numbers without using a temporary variable.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();

    printf(" Enter two numbers : ");
    scanf("%d %d", &a, &b);

    a = a + b;
    b = a - b;
    a = a - b;

    printf(" After swapping : a = %d, b = %d", a, b);

    getch();
}
```

Sample Run

Input 10 20

Output After swapping : a = 20, b = 10

07 Swap two numbers using a temporary variable

AIM To swap two numbers using a temporary variable.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b, temp;
    clrscr();

    printf(" Enter two numbers : ");
    scanf("%d %d", &a, &b);

    temp = a;
    a = b;
    b = temp;

    printf(" After swapping : a = %d, b = %d", a, b);

    getch();
}
```

Sample Run

Input 10 20

Output After swapping : a = 20, b = 10

08 Check a > b using the conditional operator

AIM To input two numbers a and b and check a > b using the conditional operator.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();

    printf(" Enter two numbers : ");
    scanf("%d %d", &a, &b);

    (a > b) ? printf("a is greater") : printf("b is greater or equal");

    getch();
}
```

Sample Run

Input 5 3

Output a is greater

09 Arithmetic and relational operations using short-name method

AIM To perform arithmetic and relational operations using the short-hand method.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b, x;
    clrscr();
    printf(" Enter two values : ");
    scanf("%d %d", &a, &b);

    printf("\n Arithmetic operations : ");

    x = a;
    x += b;
    printf("\n Addition = %d", x);

    x = a;
    x -= b;
    printf("\n Subtraction = %d", x);

    x = a;
    x *= b;
    printf("\n Multiplication = %d", x);

    x = a;
    x /= b;
    printf("\n Division = %d", x);

    printf("\n\n Relational operators : ");
    printf("\n a>b = %d", a > b);
    printf("\n a<b = %d", a < b);
    printf("\n a==b = %d", a == b);
    printf("\n a!=b = %d", a != b);
    printf("\n a>=b = %d", a >= b);
    printf("\n a<=b = %d", a <= b);

    getch();
}
```

Sample Run

Input 10 5

Output Addition = 15 | Subtraction = 5 | Multiplication = 50 | Division = 2

10 Evaluate the expression (a+b)*c

AIM To evaluate the expression (a + b) * c.

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b, c, result;
    clrscr();

    printf(" Enter a, b and c : ");
    scanf("%d %d %d", &a, &b, &c);

    result = (a + b) * c;

    printf(" Result = %d", result);

    getch();
}
```

Sample Run

Input 2 3 4

Output Result = 20

11 Equivalence of two numbers

AIM To check the equivalence of two numbers.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();

    printf(" Enter two integers a and b : ");
    scanf("%d %d", &a, &b);

    if (a == b)
        printf("\n 1. Condition (a==b) is TRUE");
    else
        printf("\n 1. Condition (a==b) is FALSE");

    if (a - b == 0)
        printf("\n 2. Condition (a-b==0) is TRUE");
    else
        printf("\n 2. Condition (a-b==0) is FALSE");

    if (b != 0)
    {
        if (a / b == 1)
            printf("\n 3. Condition (a/b==1) is TRUE");
        else
            printf("\n 3. Condition (a/b==1) is FALSE");
    }
    else
        printf("\n 3. Division by zero is not possible");

    getch();
}
```

Sample Run

Input 5 5

Output Condition (a==b) is TRUE | Condition (a-b==0) is TRUE | Condition (a/b==1) is TRUE

12 Largest of two numbers

AIM To find the largest of two numbers.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();

    printf(" Enter two numbers : ");
    scanf("%d %d", &a, &b);

    if (a > b)
        printf(" a is larger");
    else
        printf(" b is larger");

    getch();
}
```

Sample Run

Input 7 3

Output a is larger

13 Maximum of three numbers

AIM To find the maximum of three numbers.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b, c;
    clrscr();

    printf(" Enter three numbers : ");
    scanf("%d %d %d", &a, &b, &c);

    if (a > b && a > c)
        printf(" A is largest");
    else if (b > a && b > c)
        printf(" B is largest");
    else
        printf(" C is largest");

    getch();
}
```

Sample Run

Input 3 9 5

Output B is largest

14 Smallest of two numbers

AIM To find the smallest of two numbers.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int a, b;
    clrscr();

    printf(" Enter two numbers : ");
    scanf("%d %d", &a, &b);

    if (a < b)
    {
        printf(" a is smaller");
    }
    else
    {
        printf(" b is smaller");
    }

    getch();
}
```

Sample Run

Input 4 8

Output a is smaller

15 Energy bill using else-if ladder

AIM To calculate the energy bill by reading the starting and ending meter readings using an else-if ladder.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    float start, end, units, bill;
    clrscr();

    printf(" Enter starting meter reading : ");
    scanf("%f", &start);
    printf(" Enter ending meter reading : ");
    scanf("%f", &end);

    units = end - start;

    if (units > 200 && units <= 500)
    {
        bill = units * 2.50;
    }
    else if (units >= 100 && units < 200)
    {
        bill = units * 2.50;
    }
    else
    {
        bill = units * 1.50;
    }

    printf("\n Units consumed = %f", units);
    printf("\n Energy bill = Rs %f", bill);

    getch();
}
```

Sample Run

Input 1000 -> 1250

Output Units consumed = 250.000000 | Energy bill = Rs 625.000000

16 Leap year or not

AIM To check whether a year is a leap year or not.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int year;
    clrscr();

    printf(" Enter a year : ");
    scanf("%d", &year);

    if (year % 4 == 0)
    {
        printf(" Leap year");
    }
    else
    {
        printf(" Not a leap year");
    }

    getch();
}
```

Sample Run

Input 2024

Output Leap year

17 Odd or even

AIM To check whether a number is odd or even.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int a;
    clrscr();

    printf(" Enter the number : ");
    scanf("%d", &a);

    if (a % 2 == 0)
    {
        printf(" a is even");
    }
    else
    {
        printf(" a is odd");
    }

    getch();
}
```

Sample Run

Input 7

Output a is odd

18 Gross salary of an employee

AIM To calculate the gross salary of an employee.

Conditions

Basic Salary	DA	HRA	Conveyance
>= 5000	110% of basic	20% of basic	500
> 3000 & < 5000	100% of basic	15% of basic	400
<= 3000	90% of basic	10% of basic	300

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    float basic_salary, DA, HRA, conveyance, gross_salary;
    clrscr();

    printf(" Enter basic salary : ");
    scanf("%f", &basic_salary);

    if (basic_salary >= 5000)
    {
        DA = 1.10 * basic_salary;
        HRA = 0.20 * basic_salary;
        conveyance = 500.00;
    }
    else if (basic_salary >= 3000 && basic_salary < 5000)
    {
        DA = 1.00 * basic_salary;
        HRA = 0.15 * basic_salary;
        conveyance = 400.00;
    }
    else if (basic_salary >= 3000)
    {
        DA = 0.90 * basic_salary;
        HRA = 0.10 * basic_salary;
        conveyance = 300.00;
    }
    else
    {
        printf(" Invalid value");
    }

    gross_salary = basic_salary + DA + HRA + conveyance;
    printf(" Gross salary = %.2f", gross_salary);
```

```
    getch();
}
```

Sample Run

Input 6000

Output Gross salary = 14300.00

19 Average of six subjects with division classification

AIM To find the average of marks in six subjects and the result.

Conditions

Average	Result
34 & < 50	Third Division
50 & < 60	Second Division
60 & < 75	First Division
75 & 100	Distinction
Mark in any subject < 35	Fail

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    float marks[6], sum = 0, avg;
    int i = 0, has_failed = 0;
    clrscr();

    printf(" Enter the marks for 6 subjects : \n");
    do
    {
        printf(" Subject %d : ", i + 1);
        scanf("%f", &marks[i]);
        if (marks[i] < 35.0)
        {
            has_failed = 1;
        }
        sum += marks[i];
        i++;
    } while (i < 6);

    avg = sum / 6.0;

    printf("\n ... AVERAGE RESULT ... \n");
    printf(" Average mark : %.2f\n", avg);

    printf(" Result : ");
    if (has_failed)
    {
        printf(" Fail (marks in one or more subjects are less than 35)\n");
    }
    else if (avg >= 75.0 && avg <= 100.0)
    {
        printf(" Distinction\n");
    }
    else if (avg >= 60.0 && avg < 75.0)
```

```
    {
        printf(" First Division\n");
    }
    else if (avg >= 50.0 && avg < 60.0)
    {
        printf(" Second Division\n");
    }
    else if (avg >= 35.0 && avg < 50.0)
    {
        printf(" Third Division\n");
    }
    else
    {
        printf(" Fail\n");
    }

    getch();
}
```

Sample Run

Input 80,85,90,95,88,75

Output Average mark : 85.50 | Result : Distinction

20 Day of a week using switch case

AIM To print the day of a week using switch case.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int choice;
    clrscr();

    printf(" Menu\n");
    printf(" * * * * ..... \n");
    printf(" 1. Sunday\n");
    printf(" 2. Monday\n");
    printf(" 3. Tuesday\n");
    printf(" 4. Wednesday\n");
    printf(" 5. Thursday\n");
    printf(" 6. Friday\n");
    printf(" 7. Saturday\n");
    printf(" Enter your choice : ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf(" Sunday");
            break;
        case 2:
            printf(" Monday");
            break;
        case 3:
            printf(" Tuesday");
            break;
        case 4:
            printf(" Wednesday");
            break;
        case 5:
            printf(" Thursday");
            break;
        case 6:
            printf(" Friday");
            break;
        case 7:
            printf(" Saturday");
            break;
        default:
            printf(" Invalid choice");
    }
```

```
    getch();
}
```

Sample Run

Input 3

Output Tuesday

21 Menu driven arithmetic calculations using switch case

AIM To write a menu driven program to perform arithmetic calculations using switch case.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int choice;
    int x, y;
    float a, b, sum, sub, mul, div;
    clrscr();

    printf(" Menu\n");
    printf(" * * * ..... * \n");
    printf(" 1. Addition\n");
    printf(" 2. Subtraction\n");
    printf(" 3. Multiplication\n");
    printf(" 4. Division\n");
    printf(" 5. Remainder\n");
    printf(" 6. Largest of two numbers\n");
    printf(" 7. Exit\n");
    printf(" Enter your choice : ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf(" Enter two numbers : ");
            scanf("%f %f", &a, &b);
            sum = a + b;
            printf(" Addition = %.2f", sum);
            break;
        case 2:
            printf(" Enter two numbers : ");
            scanf("%f %f", &a, &b);
            sub = a - b;
            printf(" Subtraction = %.2f", sub);
            break;
        case 3:
            printf(" Enter two numbers : ");
            scanf("%f %f", &a, &b);
            mul = a * b;
            printf(" Multiplication = %.2f", mul);
            break;
        case 4:
            printf(" Enter two numbers : ");
            scanf("%f %f", &a, &b);
            div = a / b;
```

```
            printf(" Division = %.2f", div);
            break;
        case 5:
            printf(" Enter two numbers : ");
            scanf("%d %d", &x, &y);
            printf(" Remainder = %d", x % y);
            break;
        case 6:
            printf(" Enter two numbers : ");
            scanf("%d %d", &x, &y);
            if (x > y)
                printf(" Largest = %d", x);
            else
                printf(" Largest = %d", y);
            break;
        case 7:
            printf(" Exit");
            break;
        default:
            printf(" Invalid choice");
    }

    getch();
}
```

Sample Run

Input 1 -> 10 5

Output Addition = 15.00

22 Display star pattern

AIM To display the star pattern.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int i, j;
    clrscr();

    for (i = 1; i <= 5; i++)
    {
        for (j = 1; j <= i; j++)
        {
            printf("*");
        }
        printf("\n");
    }

    getch();
}
```

Sample Run

Input (none)

Output

```
*

**

***

****

*****
```

23 Display number triangle pattern

AIM To display the number triangle pattern.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int i, j;
    clrscr();

    for (i = 1; i <= 5; i++)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%d", j);
        }
        printf("\n");
    }

    getch();
}
```

Sample Run

Input (none)

Output

```
1
12
123
1234
12345
```

24 Print Floyd triangle

AIM To display the Floyd triangle.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int i, j, k = 1;
    clrscr();

    for (i = 1; i <= 5; i++)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%d ", k);
            k++;
        }
        printf("\n");
    }

    getch();
}
```

Sample Run

Input (none)

Output

```
1

2 3

4 5 6

7 8 9 10

11 12 13 14 15
```

25 Find the sum of array elements

AIM To find the sum of array elements.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int n, i, sum = 0;
    int arr[100];
    clrscr();

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
        sum += arr[i];
    }

    printf("Sum of array elements = %d", sum);

    getch();
}
```

Sample Run

Input 5 -> 1 2 3 4 5

Output Sum of array elements = 15

26 Find the largest element in an array

AIM To find the largest element in an array.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int n, i, largest;
    int arr[100];
    clrscr();

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    largest = arr[0];
    for (i = 1; i < n; i++)
    {
        if (arr[i] > largest)
        {
            largest = arr[i];
        }
    }

    printf("Largest element = %d", largest);

    getch();
}
```

Sample Run

Input 5 -> 3 9 4 7 1

Output Largest element = 9

27 Display a matrix

AIM To display a matrix.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int r, c, i, j;
    int matrix[10][10];
    clrscr();

    printf("Enter number of rows and columns: ");
    scanf("%d%d", &r, &c);

    printf("Enter elements of the matrix:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            scanf("%d", &matrix[i][j]);
        }
    }

    printf("The entered matrix is:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            printf("%d\t", matrix[i][j]);
        }
        printf("\n");
    }

    getch();
}
```

Sample Run

Input 2 3 -> 1 2 3 4 5 6

Output 1 2 3

4 5 6

28 Addition of two matrices

AIM To add two matrices.

```c
#include<stdio.h>
#include<conio.h>
void main()
{
    int r, c, i, j;
    int mat1[10][10], mat2[10][10], sum[10][10];
    clrscr();

    printf("Enter number of rows and columns: ");
    scanf("%d%d", &r, &c);

    printf("Enter elements of first matrix:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            scanf("%d", &mat1[i][j]);
        }
    }

    printf("Enter elements of second matrix:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            scanf("%d", &mat2[i][j]);
        }
    }

    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            sum[i][j] = mat1[i][j] + mat2[i][j];
        }
    }

    printf("Sum of the two matrices:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            printf("%d\t", sum[i][j]);
        }
        printf("\n");
```

```
    }

    getch();
}
```

Sample Run

Input 2 2 -> 1 2 3 4 | 5 6 7 8

Output 6 8

10 12